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System and method for priming an intravenous line

Abstract

A system for priming an intravenous line includes a closed system transfer device having a first connector and a second connector, a connector device having a first connector and a second connector, a collection device having a first connector, a cannula having a distal end, and a sleeve having a first position where the distal end of the cannula is positioned within the sleeve and a second position where the distal end of the cannula is positioned outside of the sleeve, and at least one collection container having a closure. The collection container defines a volume having a negative pressure, with the cannula of the collection device configured to pierce the closure of the collection container.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application claims priority to U.S. Provisional Application Ser. No. 62/970,226, entitled “System and Method for Priming an Intravenous Line”, filed Feb. 5, 2020, the entire disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION
Field of the Disclosure
(1) The present application relates generally to a system and method for priming an intravenous line.
Description of the Related Art

(2) Health care providers reconstituting, transporting, and administering hazardous drugs, such as chemotherapeutic drugs, can put health care providers at risk of exposure to these medications and present a major hazard in the health care environment. For example, health care providers treating cancer patients risk being exposed to chemotherapy drugs and their toxic effects. Unintentional chemotherapy exposure can affect the nervous system, impair the reproductive system, and bring an increased risk of developing blood cancers in the future. In order to reduce the risk of health care providers being exposed to toxic drugs, the closed transfer of these drugs becomes important.
(3) After connecting an intravenous line to an infusion container containing a hazardous drug using a connector, such as an IV bag spike, the intravenous line needs to be primed prior to infusion of the medicament within the infusion container into a patient. The intravenous line may be pre-primed using saline or other appropriate fluid. However, in cases where priming with a hazardous drug in line is desired, the hazardous drug is pushed through the intravenous line to remove the air with any fluid from the line being caught by a receptacle, such as a trash receptacle. Releasing hazardous drugs into an unsealed receptacle presents risks to health care providers and patients.

SUMMARY OF THE INVENTION

(4) In one aspect or embodiment, a system for priming an intravenous line includes a closed system transfer device having a first connector and a second connector, with the first connector of the closed system transfer device configured to be connected to a connector of an intravenous line, a connector device having a first connector and a second connector, with the first connector of the connector device configured to be connected to the second connector of the closed system transfer device, a collection device having a first connector, a cannula having a distal end, and a sleeve having a first position where the distal end of the cannula is positioned within the sleeve and a second position where the distal end of the cannula is positioned outside of the sleeve. The first connector of the collection device is configured to be connected to the second connector of the connector device. The system also includes at least one collection container having a closure, with the collection container defining a volume having a negative pressure. The cannula of the collection device is configured to pierce the closure of the collection container.
(5) The closed system transfer device may be a syringe adapter. The connector device may be a patient connector. The closed system transfer device may be a closeable male luer adapter. The connector device may be a needle-free connector. The collection device may be a blood collection device, and the at least one collection container may be a blood collection tube.
(6) The closed system transfer device, the connector device, the collection device, and the at least one collection container may be configured to facilitate transfer of fluid from an intravenous line through the closed system transfer device, the connector device, the collection device, and into the at least one collection container without leaking of the fluid outside of the system.
(7) The second connector of the closed system transfer device may include a membrane, and the first connector of the connector device may include a membrane, with the membrane of the second connector of the closed system transfer device configured to engage the membrane of the first connector of the connector device.
(8) In a further aspect or embodiment of the present invention, a method of priming an intravenous line includes the steps of providing an intravenous line and a closed system transfer device comprising a first connector and a second connector, and connecting the closed system transfer device to the intravenous line. The method also includes the steps of providing a connector device comprising a first connector and a second connector, providing a collection device comprising a first connector, a cannula having a distal end, and a sleeve having a first position where the distal end of the cannula is positioned within the sleeve, and a second position where the distal end of the cannula is positioned outside of the sleeve, and connecting the connector device to the collection device. The method further includes the steps of connecting the closed system transfer device to the connector device, providing a collection container comprising a closure, the collection container defining a volume having a negative pressure, and priming the intravenous line by inserting the at

least one collection container into the collection device such that the cannula of the collection device pierces the closure of the collection container.
(9) The first connector of the connector device may be configured to be connected to the second connector of the closed system transfer device, and the first connector of the collection device may be configured to be connected to the second connector of the connector device.
(10) In certain configurations, the method may also include the step of disconnecting the collection device and the connector device from the closed system transfer device.
(11) The closed system transfer device may include a syringe adapter, and the connector device may include a patient connector. Optionally, the closed system transfer device may include a closeable male luer adapter, and the connector device may include a needle-free connector. In further configurations, the collection device may include a blood collection device, and the at least one collection container may be a blood collection tube.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above-mentioned and other features and advantages of this disclosure, and the manner of attaining them, will become more apparent and the disclosure itself will be better understood by reference to the following descriptions of aspects of the disclosure taken in conjunction with the accompanying drawings, wherein:
(2) FIG. **1** is front view of a system for priming an intravenous line according to one aspect or embodiment of the present application.
(3) FIG. **2** is a front view of the system of FIG. **1**, showing a closed system transfer device being connected to an intravenous line.
(4) FIG. **3** is a front view of the system of FIG. **1**, showing a connector being secured to a transfer device.
(5) FIG. **4** is a front view of the system of FIG. **1**, showing a collection tube inserted into the transfer device.
(6) FIG. **5** is a front view of the system of FIG. **1**, showing components of the system connected to each other.
(7) FIG. **6** is a cross-sectional view taken along line **6-6** in FIG. **2**.
(8) FIG. **7** is a cross-sectional view taken along line **7-7** in FIG. **3**.
(9) FIG. **8** is a cross-sectional view taken of the collection tube and transfer device shown in FIG. **4**.
(10) FIG. **9** is a front view of a connector according to a further aspect or embodiment of the present application.
(11) FIG. **10** is a cross-sectional view taken along line **10-10** in FIG. **9**.
(12) FIG. **11** is a front view of a closed system transfer device according to a further aspect or embodiment of the present application.
(13) FIG. **12** is a cross-sectional view taken along line **12-12** in FIG. **11**.
(14) Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate exemplary aspects of the disclosure, and such exemplifications are not to be construed as limiting the scope of the disclosure in any manner.
DETAILED DESCRIPTION
(15) The following description is provided to enable those skilled in the art to make and use the described aspects contemplated for carrying out the invention. Various modifications, equivalents, variations, and alternatives, however, will remain readily apparent to those skilled in the art. Any and all such modifications, variations, equivalents, and alternatives are intended to fall within the spirit and scope of the present invention.

(16) For purposes of the description hereinafter, the terms “upper”, “lower”, “right”, “left”, “vertical”, “horizontal”, “top”, “bottom”, “lateral”, “longitudinal”, and derivatives thereof shall relate to the invention as it is oriented in the drawing figures. However, it is to be understood that the invention may assume various alternative variations, except where expressly specified to the contrary. It is also to be understood that the specific devices illustrated in the attached drawings, and described in the following specification, are simply exemplary aspects of the invention. Hence, specific dimensions and other physical characteristics related to the aspects disclosed herein are not to be considered as limiting. All numbers and ranges used in the specification and claims are to be understood as being modified in all instances by the term “about”. By “about” is meant plus or minus twenty-five percent of the stated value, such as plus or minus ten percent of the stated value. However, this should not be considered as limiting to any analysis of the values under the doctrine of equivalents.
(17) The terms “first”, “second”, and the like are not intended to refer to any particular order or chronology, but refer to different conditions, properties, or elements.
(18) Referring to FIGS. **1-12**, in one aspect or embodiment, a system **10** for priming an intravenous line **12** includes a closed system transfer device **14**, a connector device **16**, a collection device **18**, and a collection container **20**. The closed system transfer device **14** has a first connector **22** and a second connector **24**, with the first connector **22** of the closed system transfer device **14** configured to be connected to a connector **26**, such as a male luer connector, of the intravenous line **12**. The intravenous line **12** may be connected to an infusion container **28**, such as an IV bag, via a bag spike **30** or other suitable arrangement. The connector device **16** has a first connector **32** and a second connector **34**, with the first connector **32** of the connector device **16** configured to be connected to the second connector **24** of the closed system transfer device **14**. The collection device **18** has a first connector **36**, a cannula **38** having a distal end **40**, and a sleeve **42** having a first position where the distal end **40** of the cannula **38** is positioned within the sleeve **42** and a second position where the distal end **40** of the cannula **38** is positioned outside of the sleeve **42**. More specifically, the sleeve **42** is configured to be compressed when the collection container **20** is inserted into the collection device **18** to move the sleeve **42** from the first position to the second position. When the collection container **20** is removed, the sleeve **42** is configured to return to the first position to seal the cannula **38** of the collection device **18**. The first connector **36** of the collection device **18** is configured to be connected to the second connector **34** of the connector device **16**. The collection container **20** includes a closure **44**, with the collection container **20** defining a volume **46** having a negative pressure. The cannula **38** of the collection device **18** is configured to pierce the closure **44** of the collection container **20**. Although one collection container **20** is shown, the system **10** may include one or more collection containers **20**. In one embodiment or aspect, the first connector **22** of the closed system transfer device **14** is a female luer lock configured to be connected to a male luer connector **26** on the intravenous line **12**.
(19) As discussed in more detail below, the closed system transfer device **14**, the connector device **16**, the collection device **18**, and the at least one collection container **20** are configured to facilitate transfer of fluid from the intravenous line **12** through the closed system transfer device **14**, the connector device **16**, the collection device **18**, and into the at least one collection container **20** without leaking of the fluid outside of the system **10**.
(20) Referring to FIGS. **2** and **6**, in one embodiment or aspect, the closed system transfer device **14** is a syringe adapter having a cannula **52** and a seal arrangement **54** with a membrane **56**. The seal arrangement **54** is configured to move within the closed system transfer device **14** from a sealed position where a distal end **58** of the cannula **52** is sealed by the membrane **56** and a transfer position (not shown) where the distal end **58** of the cannula **52** extends through the membrane **56** to allow the cannula **52** to be placed in fluid communication with a mating connector, such as the connector device **16**. When the connector device **16** is inserted into the closed system transfer device **14**, the seal arrangement **54** is moved to the transfer position to place the cannula **52** in fluid

communication with the connector device **16**.

(21) Referring to FIGS. **3** and **7**, in one embodiment or aspect, the connector device **16** is a patient connector having a membrane **60** configured to engage the membrane **56** of the seal arrangement **54** of the closed system transfer device **14** when the connector device **16** is inserted into the closed system transfer device **14**. The first connector **32** of the connector device **16** includes a pair of arms **62** with projections **64** configured to engage the second connector **24** of the closed system transfer device **14**, which may be a portion of a housing **66** of the closed system transfer device **14**. Alternatively, or in addition to, the first connector **32** of the connector device **16** includes a proximal portion **68** of the connector device **16**, which engages a collet **70** of the closed system transfer device **14**.

(22) Referring to FIGS. **9** and **10**, in a further embodiment or aspect, the connector device **16** is a needle-free connector **72** having an elastomeric sleeve **74**, with the first connector **32** being a female luer connection and the second connector **34** being a male luer connection. The needle-free connector **72** may be the SmartSite™ available from Becton, Dickinson and Company, although other suitable needle-free connectors may be utilized.

(23) Referring to FIGS. **11** and **12**, in a further embodiment or aspect, the closed system transfer device **14** is a closed male luer **76** having a valve arrangement **78**. The first connector **22** is a female luer connection and the second connector **24** is a male luer connector configured to cooperate with the needle-free connector **72** of FIGS. **9** and **10**. The closed male luer **76** may be the Texium™ closed male luer available from Becton, Dickinson and Company, although other suitable closed male luers may be utilized.

(24) In one embodiment or aspect, the closed system transfer device **14** is formed integrally, bonded, or adhered to the intravenous line **12** or the male luer connector **26** of the intravenous line **12**. In one embodiment or aspect, the connector device **16** is formed integrally, bonded, or adhered to the collection device **18**.

(25) Referring to FIG. **8**, in one embodiment or aspect, the collection device **18** is a blood collection device and the collection container **20** is a blood collection tube, such as the BD Vacutainer® blood collection tube, although other suitable collection devices or collection containers may be utilized.

(26) Referring to FIGS. **2-5**, in one embodiment or aspect, a method of priming the intravenous line **12** using the system **10** includes: connecting the closed system transfer device **14** to the intravenous line **12** (FIG. **2**); connecting the connector device **16** to the collection device **18** (FIG. **3**); connecting the closed system transfer device **14** to the connector device **16** (FIG. **4**); and priming the intravenous line **12** by inserting the at least one collection container **20** into the collection device **18** such that the cannula **38** of the collection device **18** pierces the closure **44** of the collection container **20** (FIGS. **4** and **5**). The method may further include disconnecting the collection device **18** and the connector device **16** from the closed system transfer device **14**.

(27) The first connector **22** of the closed system transfer device **14** is connected to the male luer connection **26** of the intravenous line **12**, the second connector **34** of the connector device **16** is secured to the first connector **36** of the collection device **18**, and the second connector **24** of the closed system transfer device **14** is secured to the first connector **32** of the connector device **16**. When the collection container **20** is inserted into the collection device **18**, the collection container **20** moves the sleeve **42** from the first position to the second position with the cannula **38** piercing the closure **44** of the collection container **20** such that the cannula **38** is in fluid communication with the volume **46** having the negative pressure. The negative pressure of the collection container **20** draws air and/or fluid from the intravenous line **12**, through the closed system transfer device **14**, through the connector device **16**, through the collection device **18** and the cannula **38** of the collection device **18**, and into the collection container **20** thereby purging air from the intravenous line **12**. After disconnecting the connector device **16** and the collection device **18** from the closed system transfer device **14**, the closed system transfer device **14** may remain connected to the

intravenous line **12** to facilitate infusion into a patient. The system **10** is configured to prime the intravenous line **12** while in line with hazardous drugs while preventing leaks and/or dripping from the system **10** and the intravenous line **12**. Although one collection container **20** is shown, the priming of the intravenous line **12** and other components may require one or more collection containers **20** depending on the volume of the intravenous line **12** and other components as well as the volume **46** of the collection container **20**.

(28) While this disclosure has been described as having exemplary designs, the present disclosure can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the disclosure using its general principles. To the extent possible, one or more features of any embodiment or aspect can be combined with one or more features of any other embodiment or aspect. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

Claims

1. A system for priming an intravenous line comprising: a closed system transfer device comprising a first connector, a second connector, and a cannula extending at least partially between the first connector and the second connector, the first connector of the closed system transfer device configured to be connected to a connector of an intravenous line; a connector device comprising a first connector and a second connector, the first connector of the connector device configured to be received within and connected to the second connector of the closed system transfer device; a collection device comprising a first connector, a cannula having a distal end, and a sleeve having a first position where the distal end of the cannula of the collection device is positioned within the sleeve and a second position where the distal end of the cannula of the collection device is positioned outside of the sleeve, the first connector of the collection device configured to be connected to the second connector of the connector device; and at least one collection container comprising a closure, the at least one collection container defining a volume having a negative pressure, wherein the cannula of the collection device is configured to pierce the closure of the collection container, and wherein the second connector of the closed system transfer device comprises a membrane, wherein, in a sealed position, a distal end of the cannula of the closed system transfer device is held at least partially within the membrane of the second connector of the closed system transfer device, wherein the first connector of the connector device comprises a membrane, the membrane of the second connector of the closed system transfer device configured to engage the membrane of the first connector of the connector device, and wherein, in a transfer position, the first connector of the connector device is received within the second connector of the closed system transfer device thereby causing the cannula of the closed system transfer device to extend through at least the membrane of the second connector of the closed system transfer device to establish fluid communication between the closed system transfer device and the connector device.
2. The system of claim 1, wherein the closed system transfer device comprises a syringe adapter.
3. The system of claim 2, wherein the connector device comprises a patient connector.
4. The system of claim 1, wherein the closed system transfer device comprises a closeable male luer adapter.
5. The system of claim 1, wherein the connector device comprises a needle-free connector.
6. The system of claim 1, wherein the collection device comprises a blood collection device, and wherein the at least one collection container comprises a blood collection tube.
7. The system of claim 1, wherein the closed system transfer device, the connector device, the collection device, and the at least one collection container are configured to facilitate transfer of fluid from the intravenous line through the closed system transfer device, the connector device, and

the collection device, and into the at least one collection container.

8. The system of claim 7, wherein the closed system transfer device, the connector device, and the collection device are configured to transfer the fluid from the intravenous line into the at least one collection container without leaking.

9. A method of priming an intravenous line, the method comprising: providing an intravenous line and a closed system transfer device comprising a first connector, a second connector, and a cannula extending at least partially between the first connector and the second connector; connecting the closed system transfer device to the intravenous line; providing a connector device comprising a first connector and a second connector; providing a collection device comprising a first connector, a cannula having a distal end, and a sleeve having a first position where the distal end of the cannula is positioned within the sleeve, and a second position where the distal end of the cannula is positioned outside of the sleeve; connecting the connector device to the collection device; connecting the closed system transfer device to the connector device by securing the first connector of the connector device within the second connector of the closed system transfer device, thereby engaging a membrane of the second connector of the closed system transfer device with a membrane of the first connector of the connector device and causing the cannula of the closed system transfer device to extend through the membrane of the second connector of the closed system transfer device; providing a collection container comprising a closure, the collection container defining a volume having a negative pressure; and priming the intravenous line by inserting the collection container into the collection device such that the cannula of the collection device pierces the closure of the collection container.

10. The method of claim 9, wherein the first connector of the connector device is configured to be connected to the second connector of the closed system transfer device.

11. The method of claim 9, wherein the first connector of the collection device is configured to be connected to the second connector of the connector device.

12. The method of claim 9, further comprising: disconnecting the collection device and the connector device from the closed system transfer device.

13. The method of claim 9, wherein the closed system transfer device comprises a syringe adapter.

14. The method of claim 13, wherein the connector device comprises a patient connector.

15. The method of claim 9, wherein the closed system transfer device comprises a closeable male luer adapter.

16. The method of claim 9, wherein the connector device comprises a needle-free connector.

17. The method of claim 9, wherein the collection device comprises a blood collection device, and wherein the collection container comprises a blood collection tube.